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FILE NAVIGATE TEXT CODE ANALYZE SECTION RUN

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Files

71.mlx * x

C:\Users\ADMIN\Documents\MATLAB\71.mlx

```

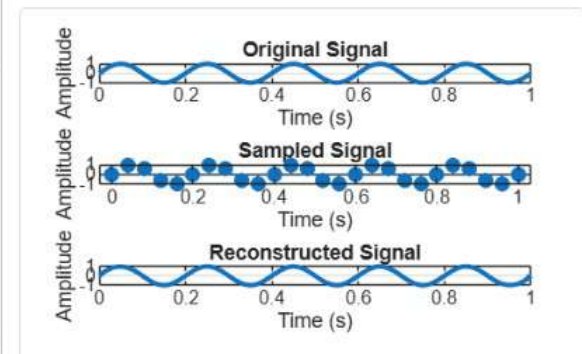
1 % Reconstruction of Signal from Samples
2 clc;
3 clear;
4 close all;
5 % Original signal frequency
6 fm = 5; % Hz
7 % Sampling frequency
8 fs = 25; % Hz
9 % Continuous-time signal
10 t = 0:0.001:1;
11 x = sin(2*pi*fm*t);
12 % Sampling
13 ts = 0:1/fs:1;
14 xs = sin(2*pi*fm*ts);
15 % Reconstruction using interpolation
16 xr = interp1(ts, xs, t, 'spline');
  
```

Workspace

Name	Value	Size	Class
fm	5	1×1	double
fs	25	1×1	double
t	1×1001 double	1×1001	double
ts	1×26 double	1×26	double
x	1×1001 double	1×1001	double
xr	1×1001 double	1×1001	double
xs	1×26 double	1×26	double

Command Window

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Files

41.mlx

61.mlx

62.mlx

71.mlx

72.mlx

Addition and multiplication...

causal non causal 42.mlx

continuous time system.mlx

Workspace

Name	Value	Size	Class
a0	0.6366	1×1	double
an	-0.0064	1×1	double
bn	-2.3733e-11	1×1	double
fs_signal	1×6284 double	1×6284	double
n	10	1×1	double
N	10	1×1	double
t	1×6284 double	1×6284	double
v	1×6284 double	1×6284	double

71.mlx

72.mlx

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```

1 % Fourier Series of Half-Wave Rectifier Output
2 clc;
3 clear;
4 close all;
5 % Time axis
6 t = 0:0.001:2*pi;
7 % Half-wave rectified signal
8 x = sin(t);
9 x(x < 0) = 0;
10 % Fourier Series approximation
11 N = 10; % Number of harmonics
12 a0 = (1/pi)*trapz(t,x);
13 fs_signal = a0/2;
14 for n = 1:N
15     an = (1/pi)*trapz(t, x.*cos(n*t));

```

Half-Wave Rectified Signal

Fourier Series Approximation

Command Window

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